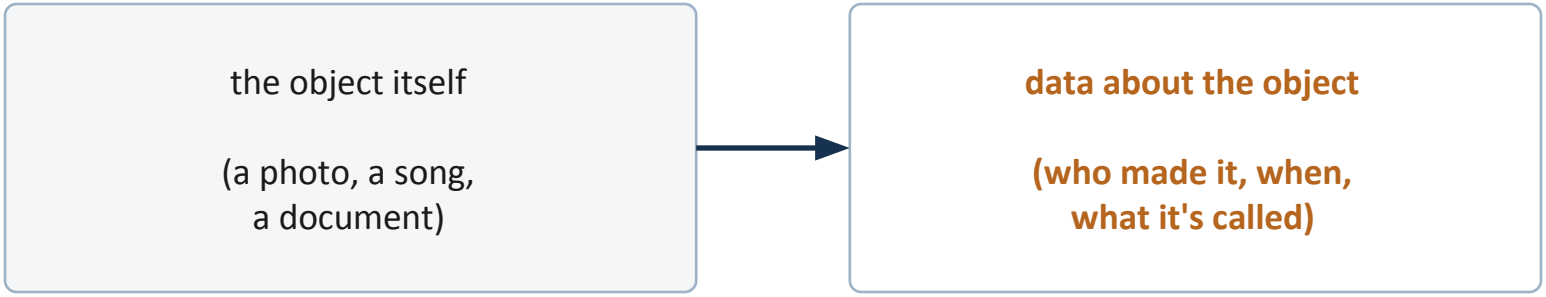
Module IV-
METADATA
Text and Multimedia Languages and Properties

The Shortcut Around the Semantic Gap

- Last topic ended on a hard problem: a computer can measure a colour histogram, but it cannot know a photo shows 'a rose'. That gap between what's measured and what's meant is the semantic gap.
- **Metadata is the decades-older, still-dominant practical answer to exactly that problem.**
- Instead of making the machine understand the content, a human describes the object once, in words — and every keyword and Boolean technique works perfectly fine on those words.
- Metadata is why multimedia search worked at all for decades before content-based feature extraction existed — and it is still how most real systems work today, underneath the fancier AI layer.

(Boolean, keyword querying) is still useful here — just pointed at a different kind of text.

What Is Metadata?



"Data about data"

- Formal definition metadata is structured information that describes the properties of a document or object — who created it, when, what it is called, what format it is in, who owns the rights to it — rather than describing its actual content.
- Crucial distinction: metadata describes the object; it is not the object.

Three Types of Metadata

Descriptive	Structural	Administrative
helps someone find it	how it's put together	technical, rights, access

- Every real metadata standard you'll meet in this topic (Dublin Core, MARC) is really just a structured way of filling in these three buckets.
- Keep this framework in mind through the rest of the lecture — every example that follows slots into one of these three.

Descriptive Metadata

- Information that helps someone find the object: title, author, subject, keywords, publication date.
- *This is the 'card catalogue' function — literally what a library index card has always contained, long before computers.*

```
Title:  Introduction to Information Retrieval

Author:  Manning, Raghavan, Schutze

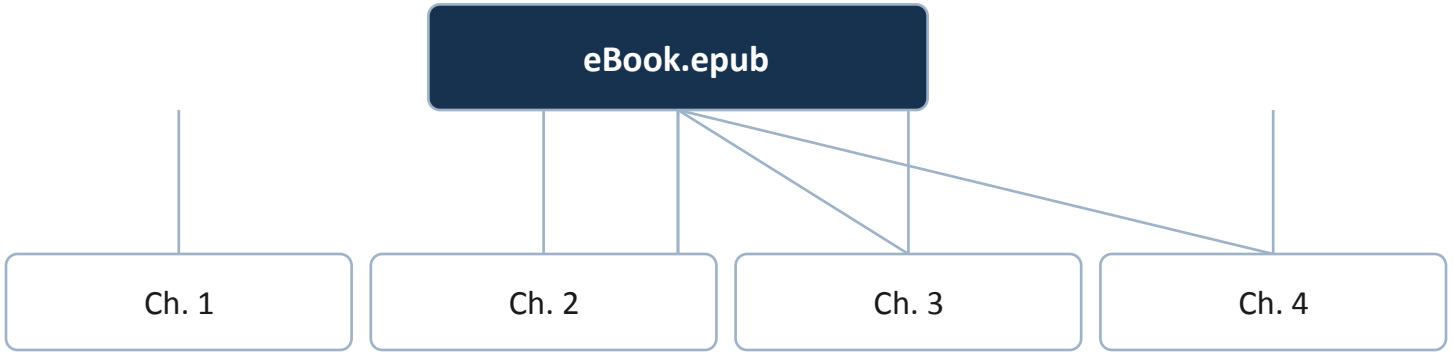
Subject:  Information Retrieval, Search Engines

Date:  2008
```

This is exactly what a library catalogue card, or a Google Books listing, shows you.

Structural Metadata

- Information about how a compound object is put together: page order in a scanned book, chapter divisions in an e-book, which images belong to which article.
- *This is the same 'Structure' dimension already taught in Multimedia IR — metadata is often exactly how that structure gets recorded and stored.*



Without this metadata, an e-reader wouldn't know which page comes after which, or where Chapter 2 begins.

Administrative Metadata

- Technical, rights, and preservation information: file format, resolution, creation date, copyright owner, access permissions.
- *This is what lets a system know how to open a file and who is allowed to see it — independent of what the file is actually about.*

Format: JPEG, 4032×3024

Created: 10 Sept 2026, 14:32

Size: 3.2 MB

Rights: © All rights reserved

This is the exact information shown in a file's 'Properties' or 'Get Info' panel on any computer.

Dublin Core

- **The single most important metadata standard to know by name.**
- Deliberately designed to be simple enough to apply to any kind of resource at all — a book, a photo, a video, a web page — unlike older, far more rigid cataloguing formats built for one specific media type.
- Originally defined as 15 core elements, all optional and all repeatable, which is exactly why it spread so widely: any system, however small, could adopt it without heavy training.
- *Named after Dublin, Ohio, USA — where the first workshop that created it was held in 1995, not Dublin, Ireland.*

The 15 Dublin Core Elements

Title	Creator	Subject
Description	Publisher	Contributor
Date	Type	Format
Identifier	Source	Language
Relation	Coverage	Rights

Every one of these is optional and repeatable — that flexibility is exactly why the standard is so widely adopted.

Real Life: Dublin Core

Thesis Repository Record

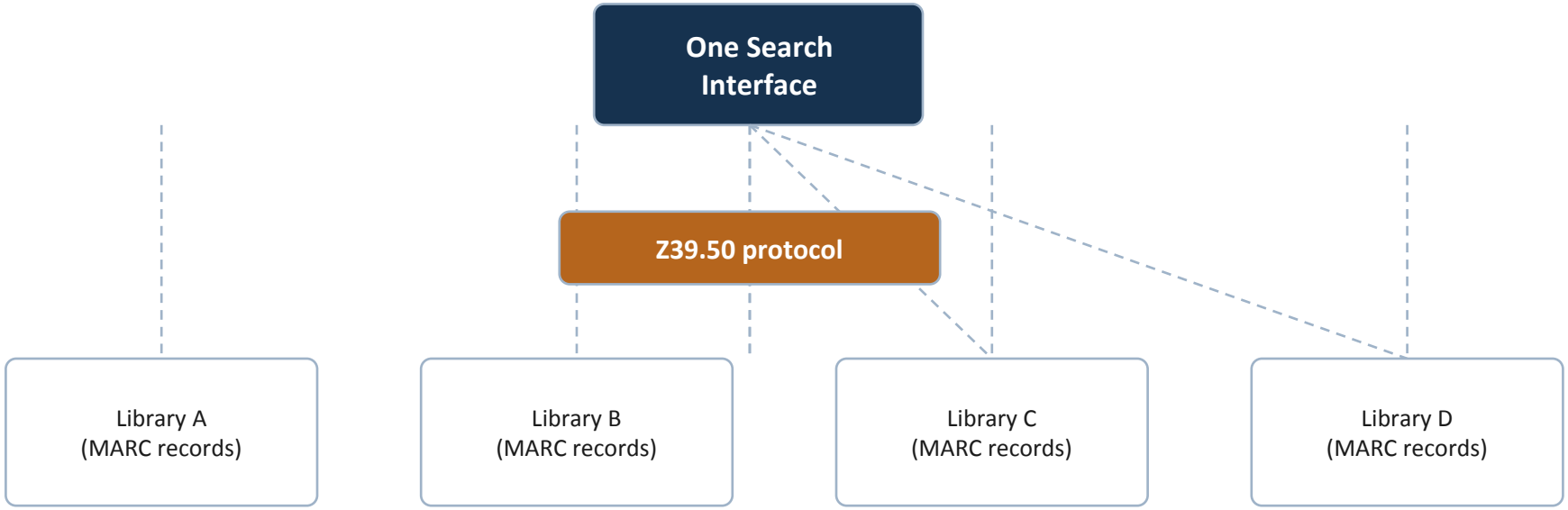
dc.title	A Study of Retrieval Models for Regional Languages
dc.creator	Sharma, A.
dc.subject	Information Retrieval; NLP
dc.date	2025-03-14
dc.rights	CC BY-NC 4.0

Most university thesis and research repositories (built on platforms like DSpace) tag every submission exactly this way.

MARC

- **MAchine-Readable Cataloging.**
- The older, far more detailed metadata format used specifically by libraries — it predates Dublin Core by decades and is much more rigid and field-heavy by comparison.
- Every physical book in a library's catalogue is described by a MARC record: hundreds of possible fields covering author, title, edition, physical description, subject headings, and more.
- **Direct callback to a topic already taught:**
  - when Z39.50 was covered as the real-world example of a Query Protocol, MARC records are exactly the bibliographic metadata being transmitted between library catalogues through that protocol.

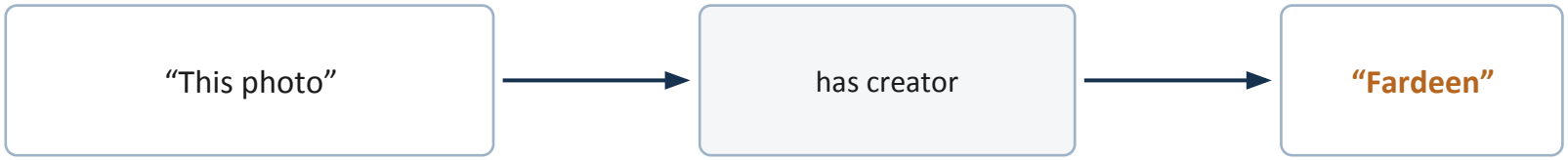
MARC Records, Travelling Over Z39.50



The protocol carries the query and the results; MARC is the metadata format those results are actually written in.

RDF

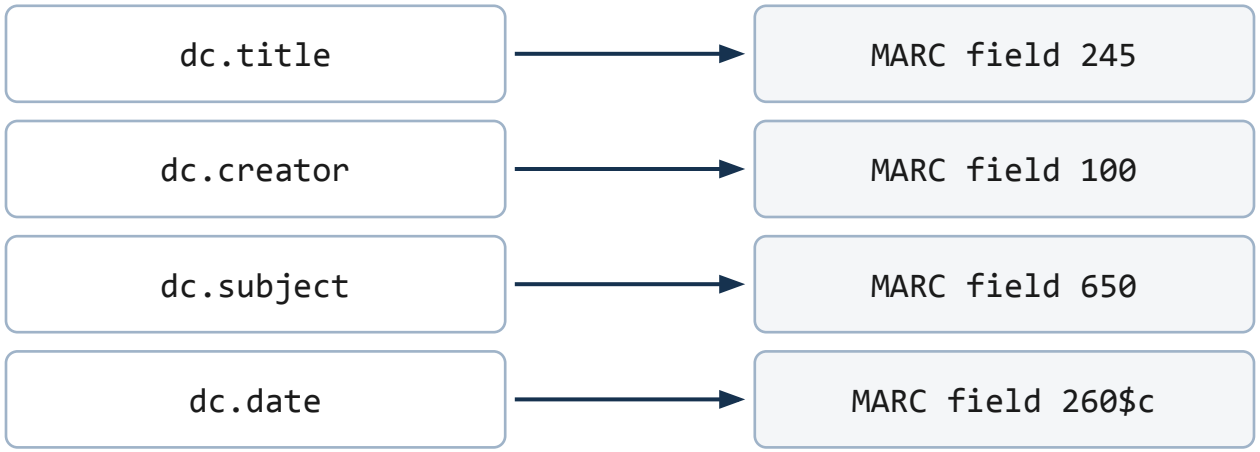
- **Resource Description Framework.**
- A more general, machine-readable way of expressing metadata — not as a fixed list of fields like Dublin Core, but as simple statements of the form subject → predicate → object.
- Designed so that metadata created by completely different, unrelated systems can still be combined and cross-referenced automatically.
- *This is the foundational idea behind what is often called the 'Semantic Web'.*



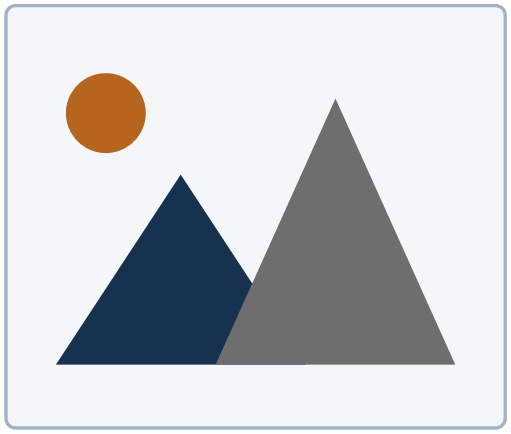
subject → predicate → object

Crosswalks

- No two institutions ever agree on exactly one metadata standard — a library might use MARC, a digital repository might use Dublin Core, a government archive might use yet another.
- **A crosswalk is a mapping between two different metadata standards, so a record described in one system can be automatically translated into another.**



Real Life: EXIF Metadata



Camera: iPhone 15 Pro

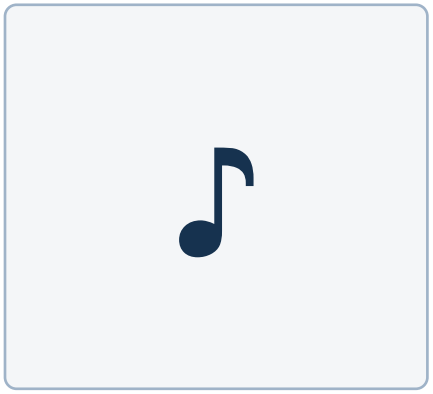
Date: 10 Sept 2026, 6:14 PM

GPS: 18.52°N, 73.85°E

Exposure: 1/250s, f/1.8

Embedded automatically, with zero effort from you — this is exactly why Google Photos can show 'Goa, December 2024' without reading a single pixel.

Real Life: ID3 Tags



Artist: Arijit Singh

Album: Tum Hi Ho

Genre: Bollywood, Romance

Track: 3 of 12

This is exactly why Spotify or iTunes can search and sort your entire library by artist, without ever 'listening' to the audio.

Real Life: HTML Meta Tags



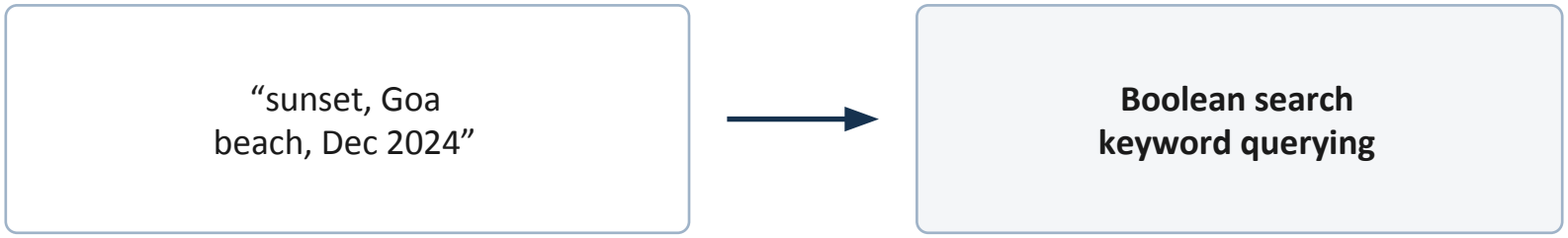
MU Info Retrieval
Syllabus 2026

mu.ac.in

```
<meta name="description" ...>
```

That preview card — title, description, thumbnail — is built entirely from the page's metadata, not from anything reading the page.

The Insight That Ties This Together



- **Metadata doesn't replace anything taught this semester — it's a translation layer.**
- Once a human writes a caption, a title, or a set of tags as metadata for a photo, video, or song, that object becomes searchable using the exact same Boolean and keyword techniques taught in week one.
- *Metadata is how multimedia retrieval quietly becomes text retrieval again.*